

Shahjalal University of Science and Technology
Department of Computer Science and Engineering



SUST Research Hub

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4th year, 2nd Semester

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Dr. Md. Forhad Rabbi, SMIEEE

Professor

Department of Computer Science and Engineering

19th July 2026

SUST Research Hub



A Project submitted to the Department of Computer Science and Engineering, Shahjalal University of Science and Technology, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering

By

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Recommendation Letter from Project Supervisor

The project entitled ***SUST Research Hub*** submitted by the students

1. A S M Sharif Mahmud Sazid
2. Sharmin Akther

is under my supervision. I, hereby, agree that the project can be submitted for examination.



Signature of the Supervisor:

Name of the Supervisor: Dr. Md. Forhad Rabbi

Date: 19th July 2026

Certificate of Acceptance of the Project

The project entitled ***SUST Research Hub*** submitted by the students

1. A S M Sharif Mahmud Sazid
2. Sharmin Akther

on 19th July 2026, hereby, accepted as the partial fulfillment of the requirements for the award of their Bachelor Degrees.



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ABSTRACT

The SUST Research Hub is a comprehensive digital platform designed to transform how academic research — including theses, publications, projects, datasets, and machine learning models — is stored, discovered, and managed within Shahjalal University of Science and Technology (SUST). It replaces the traditional practice of archiving research outputs as physical hardcopies, which inherently makes them difficult to search, verify for originality, and build upon. Traditional, paper-based or scattered digital submission processes often result in disorganized archives, inconsistent approval workflows, and limited accessibility for students, supervisors, and administrative staff. These challenges reduce research visibility, complicate the validation of new thesis topics, and hinder institutional knowledge preservation.

This project presents the design, development, and complete implementation of a modernized Thesis and Research Repository Management System. Developed as a full-scale web application, the system successfully streamlines the end-to-end lifecycle of thesis/publication/project submission, storage, supervisor approval, and public discovery across the departments of SUST.

The fully implemented platform features robust foundational and advanced capabilities, including a secure role-based access control (RBAC) system for students, supervisors, and administrators, accompanied by custom session management and automated email notifications. It encompasses complete supervisor–student workflows, enabling seamless thesis/project submission pipelines, iterative review systems, and direct feedback mechanisms. Furthermore, the repository has been expanded beyond traditional documents to fully support the hosting and management of multi-author publications, research datasets, and machine learning models, complete with popularity tracking and analytics.

A responsive, component-based frontend architecture ensures high-performance accessibility across all devices. This report documents the problem context, system requirements, architectural design, implemented modules, and testing phases. By adhering to modern web development best practices, the SUST Research Hub delivers a secure, scalable, and fully functional academic repository that actively preserves institutional knowledge and fosters future research collaboration.

Keywords: Thesis Repository, Research Management, Academic Platform, Web Application, Next.js, PostgreSQL, Role-Based Access Control, Publications, Projects, Machine Learning Models, Datasets.

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Finally, we would like to acknowledge the open-source community whose tools, libraries, and documentation—including Next.js, React, Tailwind CSS, and PostgreSQL—greatly facilitated development and learning.

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CHAPTER 1

INTRODUCTION

1.1 Background

Universities produce a large volume of academic research in the form of undergraduate theses, projects, and faculty-led studies. These research outcomes are valuable resources, yet in most institutions, including SUST, they are traditionally stored as physical hardcopies in departmental libraries, making them difficult to access and underutilized. Students often struggle to explore previous work, supervisors cannot easily verify whether a proposed topic has already been completed, and opportunities to extend or improve existing research are frequently missed.

A centralized digital research repository provides a modern alternative by supporting long-term preservation, easy academic reference, and verification of new research topics through open access to structured metadata, full-text documents, and search tools.

The SUST Research Hub is designed as a modern web-based platform that meets these institutional needs and has grown into a research ecosystem that covers theses, publications, projects, datasets, and machine learning models. By adhering to rigorous engineering standards, the platform successfully establishes a robust, highly accessible, and fully integrated environment that transforms how university research is preserved, discovered, and built upon.

1.2 Problem Statement

Traditional thesis submission and storage processes face several challenges:

- Research is stored only as physical hardcopies in libraries
- Limited accessibility to past research
- No simple way or digital solution available for students and supervisors to access previous research
- Inefficient verification of topic originality for new research
- No administrative tools for monitoring research activity
- Supervisors cannot efficiently check if a topic has already been completed
- Limited ability for new researchers to build upon existing work.

These issues highlight the need for a secure, centralized and scalable digital platform.

1.3 Objectives

The objective of this project is to design and develop the SUST Research Hub — a centralized digital research repository for academic institutions. The key objectives are: The key objectives of the proposed system are:

1. Digitize and Centralize Research Assets

- Provide a structured, secure, and searchable digital archive for all departmental theses, projects, publications, datasets, and models.
- Eliminate dependency on physical library visits.

2. Improve Research Accessibility

- Allow students and supervisors to browse, search, and review previously completed work across all content types.
- Enable keyword-based, field-based, year-based, and supervisor-based filtering.

3. Prevent Topic Duplication

- Allow supervisors and students to verify previously completed research.
- Support informed topic selection and academic originality.

4. Enable Research Continuity

- Allow students to build upon existing works.
- Provide institutional memory across multiple student batches.

5. Build a Scalable Foundation for a Full Academic Repository

- Structured user onboarding and role management.
- Secure, revocable, database-backed session authentication.
- Store all core data in a scalable relational database.
- Support collaboration through team workspaces and multi-author records.

1.4 Scope of Current Work

This revised report documents the system as of the second development phase. The system now includes:

- Full frontend interface across all content types (theses, publications, projects, datasets, models)
- User registration with admin approval
- Secure, session-based login (bcrypt password hashing; stateful DB sessions, not JWT)
- Email notifications (Resend) for registration requests and approvals
- Cloudinary integration for file uploads
- A production PostgreSQL database (Neon) covering research content, workflow, and system tables
- Public browsing and searching of published theses, with early support for browsing datasets and models
- Team-based project collaboration (teams, team_members) and multi-author thesis/publication support
- A supervisor review workflow with structured feedback and revision requests
- Full notification system by unified in-app notification center with read/unread state across all event types

CHAPTER 2

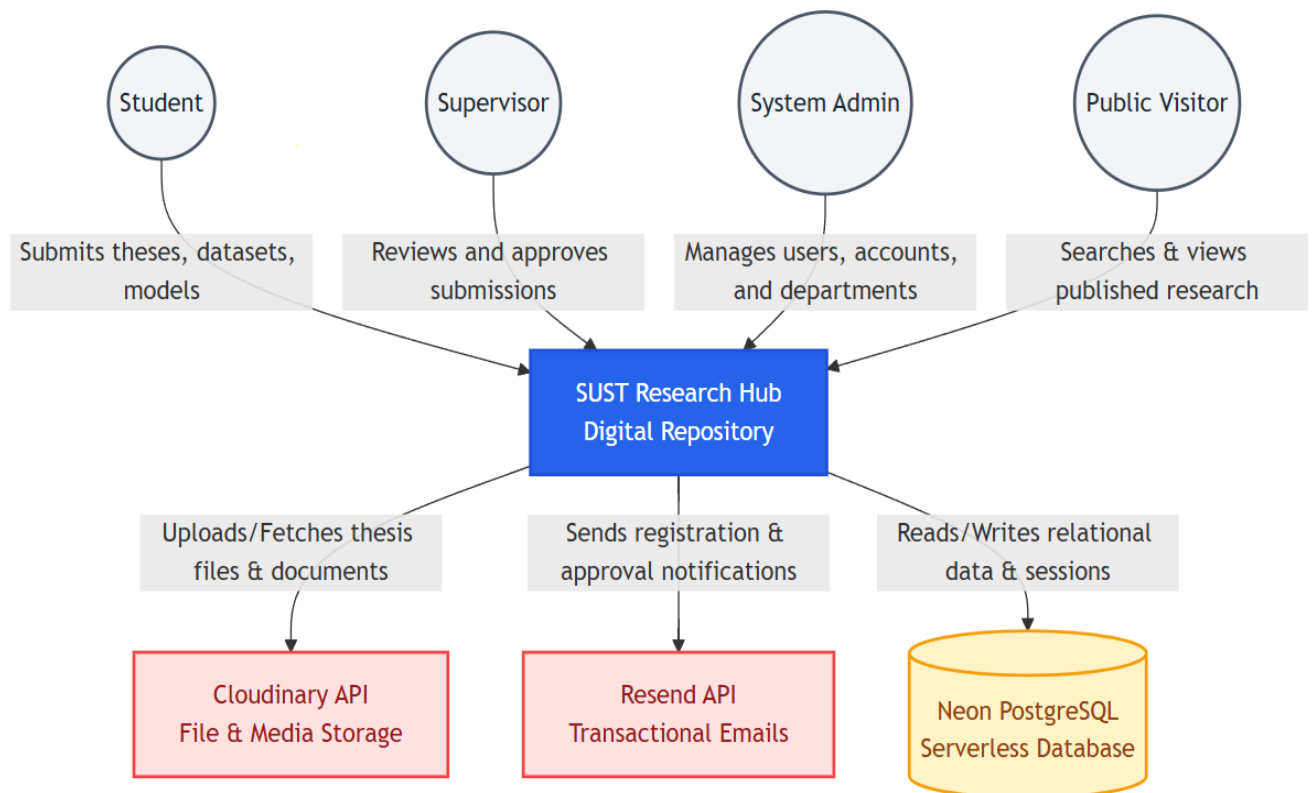
SYSTEM OVERVIEW

2.1 Overview

The SUST Research Hub is engineered as a modular, scalable web application supporting the lifecycle of academic research — no longer limited to theses. This chapter provides a high-level overview of the system, including context, actors, architecture, and technology.

2.2 System Context Overview

The SUST Research Hub operates as a centralized digital platform connecting students, supervisors, and administrators, now spanning five content types: theses, publications, projects, datasets, and models.



2.3 Actors and Roles

1. Students

- Register and await approval
- Browse and search published theses, publications, datasets, and models
- Create/join teams, submit theses and projects, request supervision

2. Supervisors

- Register and await approval
- Browse research for originality checks
- Supervise student teams, review submissions, publish/reject theses

3. Administrators

- Review and approve user registrations
- Manage system data, users, and departments
- View basic platform analytics

4. Public Users (Unauthenticated)

- View publicly available information
- Browse published theses, publications, datasets, and models

2.4 System Features

Implemented

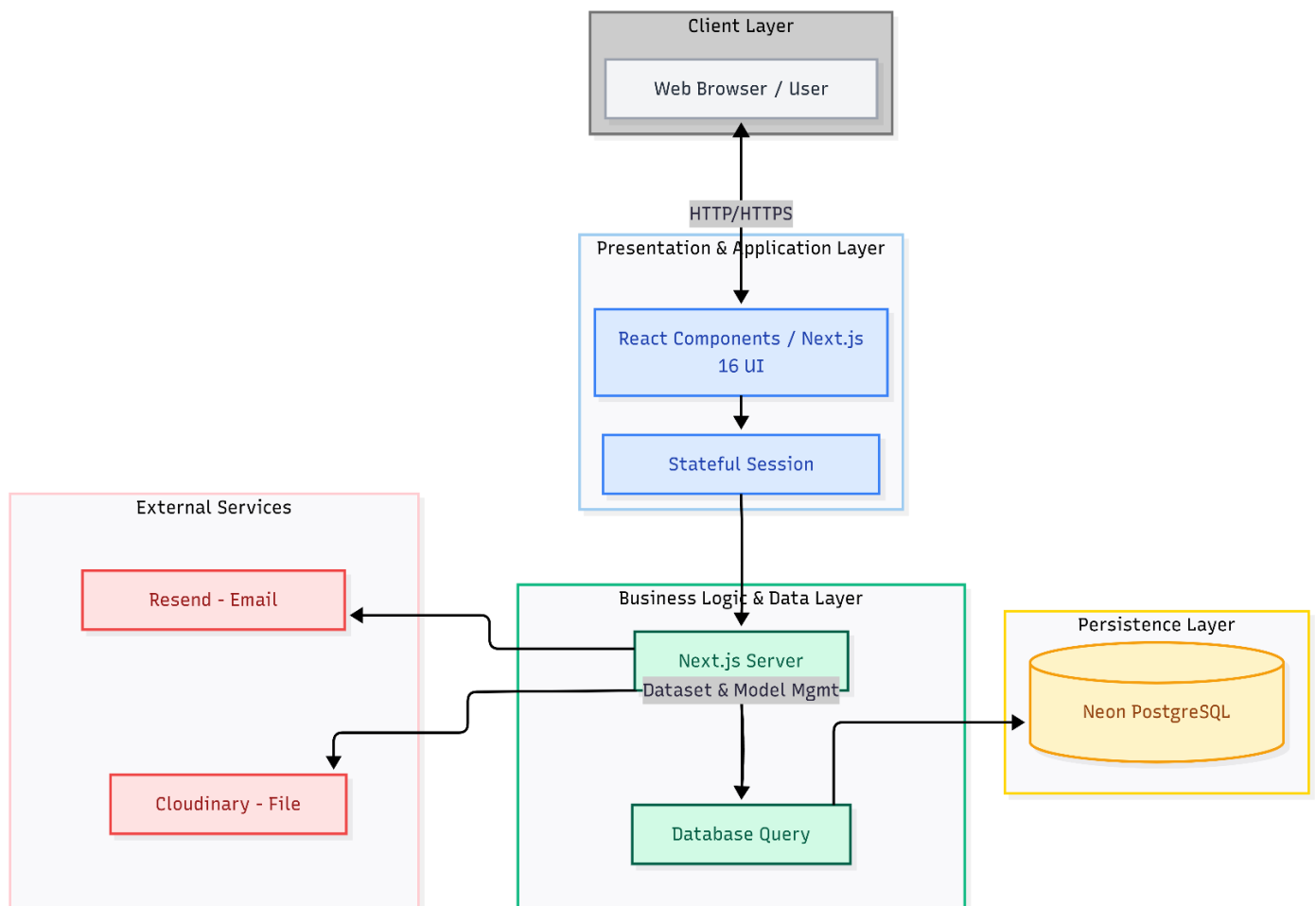
- Role-based user registration (Student, Supervisor, Admin)
- Admin approval workflow
- Secure login with bcrypt password hashing and stateful DB sessions
- Email notifications for registration and approval (Resend)
- Cloudinary-backed file uploads
- Real database integration with PostgreSQL (Neon)
- Browse and search interface for published theses
- Publications: creation and linking to theses, with citation metadata
- Projects: team-based workspaces with status tracking
- Datasets: repository entries with type/domain metadata

- Models: repository entries with framework/task metadata
- Thesis submission and supervisor review workflow, with structured feedback and revision requests
- Team collaboration via invitations (teams, team_members)
- Full notification center with read/unread state across all event types
- Responsive, professionally designed front-end

Planned for Future Phases

- File versioning and fine-grained access control on uploads
- Deeper analytics dashboards (department-wise output, supervisor activity trends)
- Smarter search (beyond keyword/filter matching), citation export, duplicate/topic detection

2.5 System Architecture



2.6 Technology Stack Summary

Layer	Technologies
Frontend	Next.js 16, React 19, TypeScript, Tailwind CSS, shadcn/ui
Backend/Server	Next.js Server Actions, bcrypt (password hashing), stateful DB sessions (session_token cookie)
File Storage	Cloudinary (files, images)
Database	Neon PostgreSQL, composite indexes for search performance
Email	Resend (registration/approval notifications)

2.7 Key System Characteristics

- **Scalable** Designed to accommodate a growing user base and content volume with efficient, indexed queries. No formal large-scale load testing has been performed; this is a design goal, not a measured guarantee.
- **Secure** Implements password hashing, role-based access, controlled user onboarding, and revocable server-side sessions.
- **Extensible** The addition of Publications, Projects, Datasets, and Models without restructuring the core thesis and user tables demonstrates the schema's extensibility in practice, not just in intent.
- **Accessible & User-Centric** The front-end is responsive and usable on mobile and desktop devices. No formal WCAG accessibility audit has been conducted.

CHAPTER 3

REQUIREMENTS ANALYSIS

3.1 Overview

This chapter outlines the functional and non-functional requirements derived from discussions with students, supervisors, and administrative staff. The goal was to understand how theses/projects are currently stored and accessed, and how a digital alternative should operate.

3.2 Stakeholders & User Classes

User Class	Description
Student	Submits research, forms teams, collaborates, browses repository across all content types
Supervisor	Reviews research, approves teams, provides feedback, publishes/rejects theses
Administrator	Approves registrations, manages users, departments, and system data
Public User	Browses published research without authentication

3.3 Functional Requirements

3.3.1 User Registration & Authentication

- FR-01 The system shall allow users to register with email, name, department, and role.
- FR-02 The system shall validate that the email is unique.
- FR-03 The system shall hash all passwords (bcrypt).

- FR-04 The system shall authenticate users via a database-backed session token, not a JWT.
- FR-05 The system shall deny login for unapproved users.
- FR-06 The system shall display meaningful error messages.
- FR-07 The system shall log all login attempts.
- FR-08 The system shall allow admins to reset user accounts.

3.3.2 Admin Approval Workflow

- FR-09 The system shall notify admins of new registration requests.
- FR-10 The system shall allow admins to review pending registrations.
- FR-11 The system shall allow admins to approve or reject accounts.
- FR-12 Approved users shall automatically gain login access.
- FR-13 Users shall be notified via email about approval/rejection.
- FR-14 Admins may attach notes to rejections.

3.3.3 User Profile Management

- FR-15 Users shall be able to view and update their profiles.
- FR-16 Only admins may change user roles.

3.3.4 Research Repository (Public Access)

- FR-17 The system shall list all published theses publicly.
- FR-18 The system shall filter theses/projects by title, supervisor, department, and year.
- FR-19 The system shall support keyword search.
- FR-20 Each thesis page shall display metadata and increment its view count on access.

3.3.5 Team Collaboration

- FR-21 Students shall be able to create teams and send collaboration invitations.

- FR-22 Invitees shall receive notifications and may accept or reject invitations.
- FR-23 Teams may edit project details collaboratively.

3.3.6 Thesis Submission & Review

- FR-24 Teams shall upload thesis files (via Cloudinary).
- FR-25 Supervisors shall review submissions and provide structured feedback.
- FR-26 Students shall be able to resubmit updated files after revision requests.
- FR-27 Supervisors shall publish or reject final theses.

3.3.7 Notification System

- FR-28 The system shall notify users of key workflow events.
- FR-29 Notifications shall be stored persistently and markable as read.

3.3.8 Administrative Tools

- FR-30 Admins shall manage users, departments, and thesis metadata.
- FR-31 Admins shall have access to basic platform analytics.

3.3.9 Publications Management

- FR-32 The system shall allow supervisors/students to create a publication record linked to one or more theses.
- FR-33 Publication records shall store DOI, ISSN, and venue metadata.
- FR-34 The system shall support multiple authors per publication.
- FR-35 Published records shall be publicly browsable and searchable.

3.3.10 Research Projects

- FR-36 The system shall allow users to create a project workspace distinct from thesis authorship.
- FR-37 Projects shall support a status field (e.g., active, completed) and multiple collaborators.
- FR-38 Projects may be linked to related theses, datasets, and models.

3.3.11 Datasets Repository

- FR-39 The system shall allow users to submit a dataset entry with type/domain metadata.
- FR-40 Datasets shall be filterable by type and domain.
- FR-41 Dataset entries shall track associated file metadata.

3.3.12 ML Models Repository

- FR-42 The system shall allow users to submit a model entry with framework and task metadata.
- FR-43 Models shall be filterable by framework and task.
- FR-44 Model entries shall track associated file metadata.

3.3.13 Database Requirements

- FR-45 The system shall store all referenced entities with enforced relational integrity.
- FR-46 Timestamps shall be auto-maintained; indexes shall support query optimization.

3.4 Non-Functional Requirements

3.4.1 Performance

Measured on the development environment (see Chapter 6 for methodology); not independently load-tested at scale.

- NFR-01: Home page load ~200–500ms
- NFR-02: Browse/search response ~150–300ms
- NFR-03: Average API response < 400ms

3.4.2 Security

- NFR-04: Password hashing via bcrypt

- NFR-05: Role-based access control (RBAC) enforced server-side
- NFR-06: Failed login attempts logged
- NFR-07: Sessions are server-revocable (DB-backed, 7-day expiry, HTTP-only cookie)

3.4.3 Usability

- NFR-08: UI shall be mobile-responsive
- NFR-09: Navigation shall be consistent across modules

3.4.4 Reliability

- NFR-10: All errors recorded server-side

3.4.5 Maintainability

- NFR-11: Modular design; normalized schema (3NF)

3.4.6 Portability

- NFR-12: Deployable on any Node.js-compatible host

3.5 External Interface Requirements

- UI built in React/Next.js
- PostgreSQL (Neon) as the database
- Resend as the email provider
- Cloudinary as the file storage provider

3.6 System Constraints

- Depends on internet connectivity
- Email and file storage are on free-tier plans with associated limits (see Ch. 7.3)

3.7 Assumptions & Dependencies

- Accurate user-submitted information
- Active admin oversight for approvals
- Supervisor departmental association is accurate at registration

SYSTEM DESIGN

4.1 Design Methodology

The system design follows IEEE 42010 architectural principles and modern industry practice. A modular, layered architecture ensures scalability, maintainability, and clear separation of concerns, and has proven flexible enough to absorb the Publications, Projects, Datasets, and Models modules without restructuring the core schema.

4.2 System Architecture View

4.2.1 Logical Architecture

- Presentation Layer: UI components (Next.js, React)
- Application Layer: Server Actions, business logic
- Data Access Layer: database queries, indexing
- Persistence & Storage Layer: PostgreSQL (Neon) + Cloudinary

4.2.2 Architectural Style

- Client–Server architecture
- Modular monolith (potential future evolution toward services)
- Server Actions as the primary backend mechanism, with limited REST endpoints where needed

4.3 Use Case Model

4.3.1 Detailed Use Cases

UCo1–UCo5 (Registration, Admin Approval, Login, Research Publish, Browse Research) are unchanged from v1. New use cases added in this phase:

UCo6: Create Project

- Actor: Student / Supervisor
- Precondition: User is logged in
- Flow: Create project → add collaborators → set status → optionally link theses/datasets/models

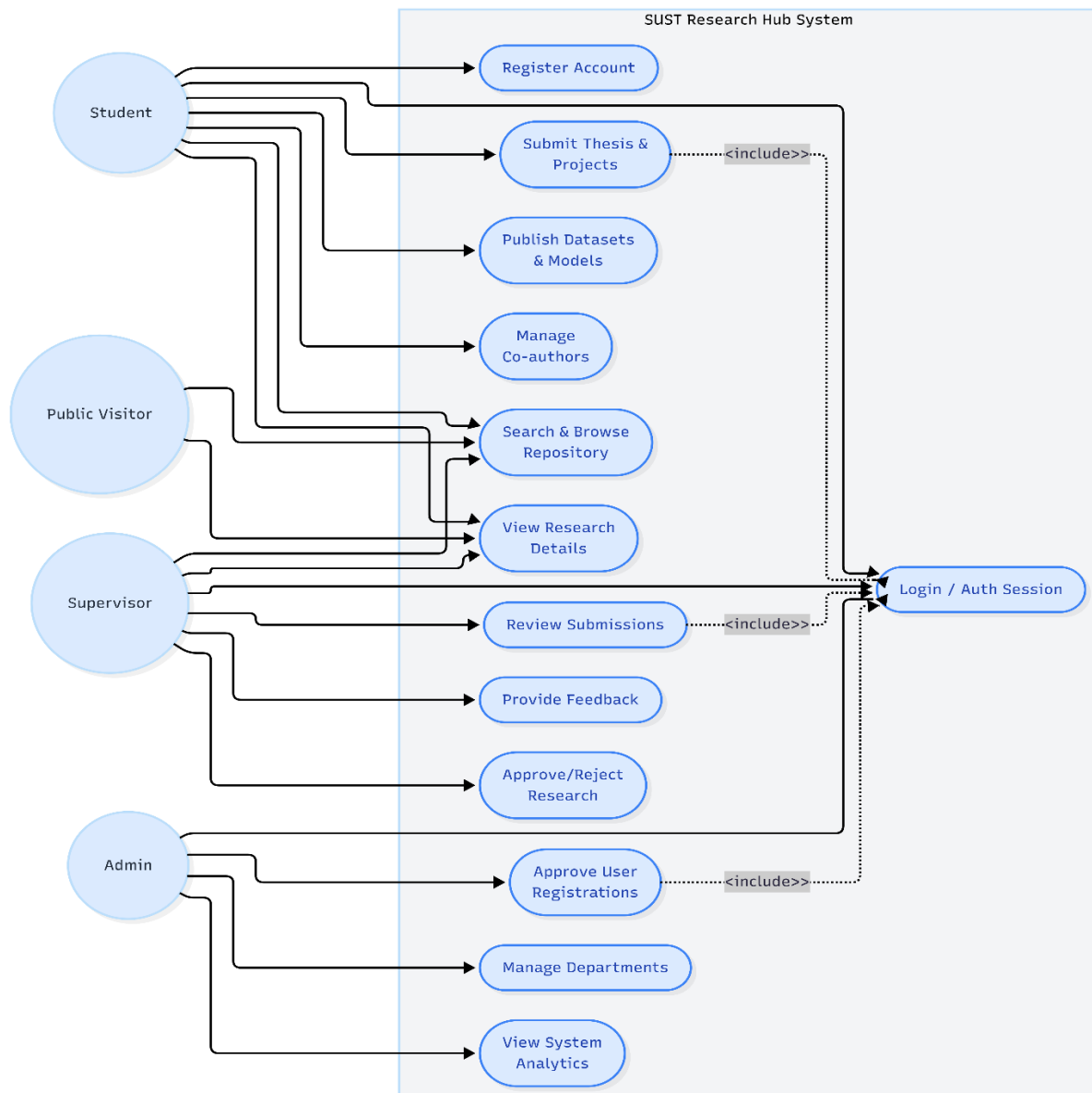
UCo7: Submit Publication

- Actor: Student / Supervisor
- Precondition: A related thesis exists (optional) or standalone publication
- Flow: Enter metadata (DOI/ISSN/venue) → add authors → publish

UCo8: Browse Datasets / Models

- Actor: Public / Student / Supervisor
- Flow: Open datasets or models page → apply type/framework filters → view detail → download

4.3.2 Use Case Diagram

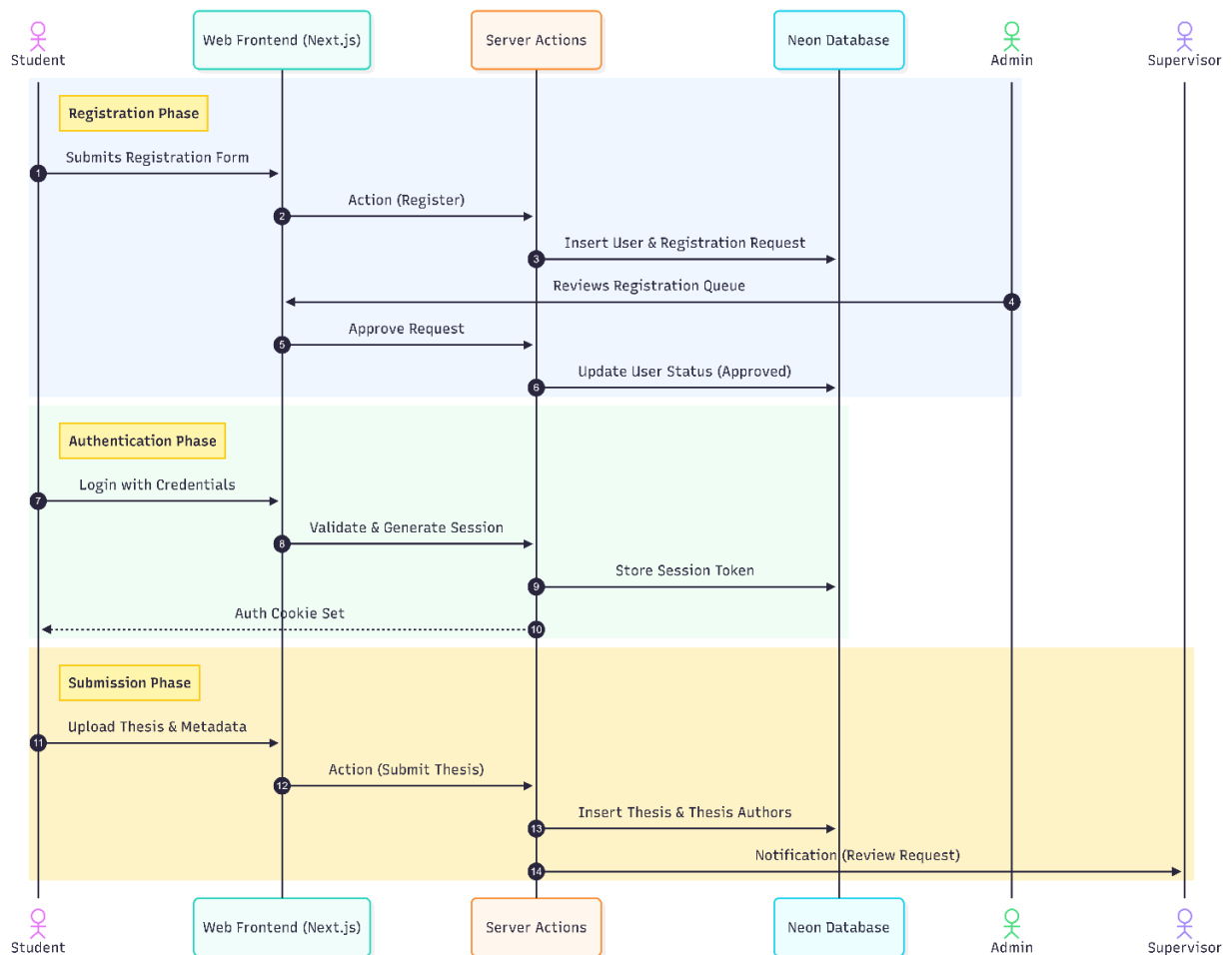


4.4 System Workflow

4.4.1 Registration & Login Workflows : registration → admin review → approval/rejection → email notification; login → credential check → bcrypt verification → session token issued → cookie set → dashboard redirect.

4.4.2 Research Publish Workflow : Team collaboration → create workspace → upload files (Cloudinary) → request supervisor → supervisor review → feedback or approval → publish → update search index.

4.4.3 Dataset / Model Submission Workflow : User submits dataset/model metadata and files → stored with type/framework tags → becomes searchable/filterable in the respective repository → optionally linked to a project or thesis.



4.5 Database Design

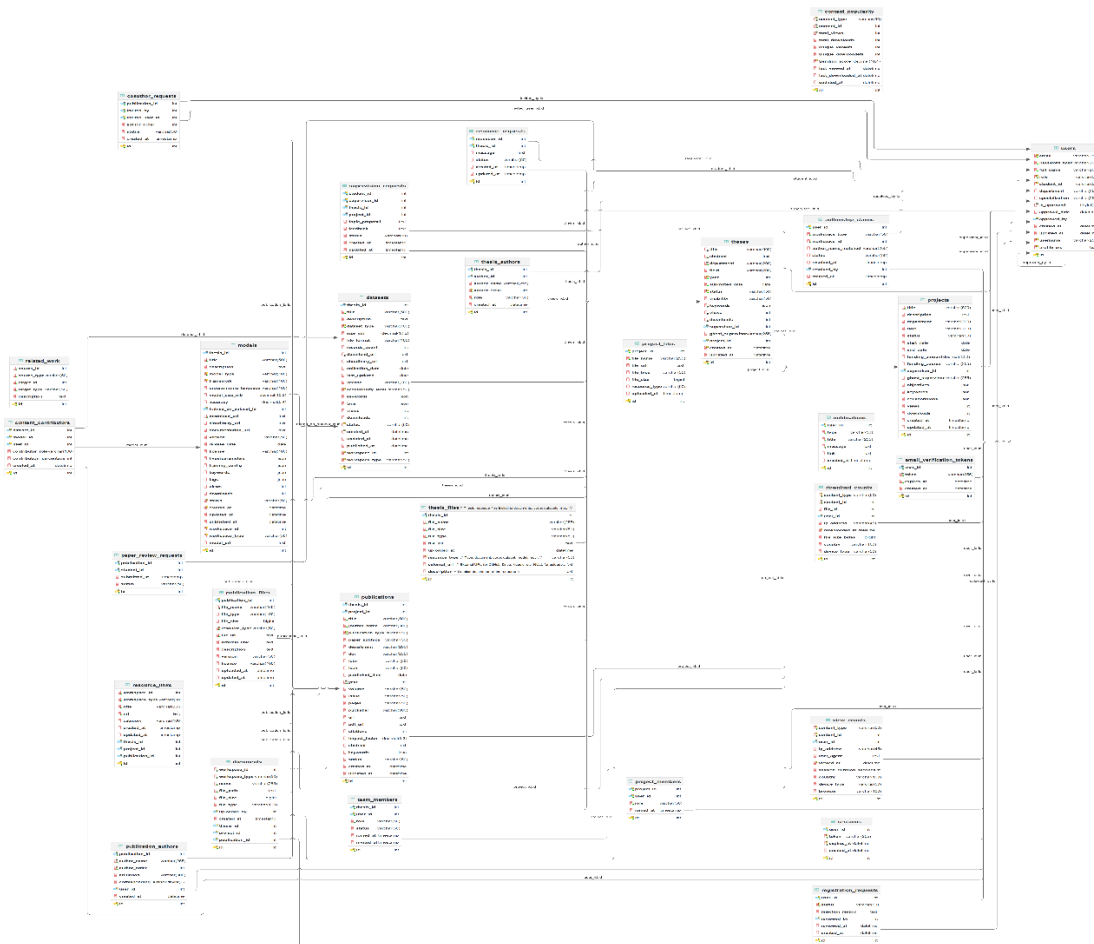
The database follows a modular, 3NF-normalized schema with UUID primary keys, foreign key constraints, and comprehensive indexing. The schema has grown from the original 13 core tables to cover Publications, Projects, Datasets, Models, and finer-grained workflow tracking.

4.5.1 Table Groups (current)

Table 4.1: Database Table Groups

Group	Tables
Core	users, departments, registration_requests, sessions
Research Content	theses, thesis_files, thesis_authors, thesis_keywords, publications, publication_authors, projects, project_collaborators, datasets, dataset_files, models, model_files, content_popularity, content_contributors
Workflow	submissions, reviews, feedback_items, revision_requests, teams, team_members
System	notifications, audit_logs, settings

4.5.2 Entity-Relationship Diagram



4.5.3 Normalization The schema maintains 1NF (atomic attributes), 2NF (UUID primary keys avoid partial dependencies), and 3NF (no transitive dependencies) across the expanded table set.

4.5.4 Indexing Strategy

- `idx_users_email` — login lookup
- `idx_theses_status`, `idx_theses_year` — public browsing/filtering
- `idx_thesis_keywords_keyword` — keyword search
- Equivalent indexes recommended on `datasets.type` and `models.framework` for the new modules, if not already present — verify against the schema

4.6 Security Design

4.6.1 Authentication

Authentication is session-based. On login, a 32-byte cryptographically secure random token is generated and stored in the sessions table with a 7-day expiry, mapped to the user's ID. The token is returned to the client in an HTTP-only session token cookie. Every authenticated request is validated with a database join between users and sessions, checking existence and expiry before the request proceeds. This design is deliberate: it keeps sessions server-revocable and avoids the refresh-token complexity of stateless JWTs, at the cost of a database lookup per authenticated request — an acceptable trade-off at the current scale.

4.6.2 Authorization (RBAC)

- Students: access to browse, submission, and personal dashboard views
- Supervisors: access to supervision and review features
- Admins: full user and content management

4.6.3 Threat Mitigation

- SQL injection prevention via parameterized queries
- Audit logs for accountability
- Account approval gating prevents unverified self-registration from gaining access
- Passwords never stored in plaintext (bcrypt)

No formal penetration test, dependency audit, or third-party security review has been performed.

4.7 UI/UX Design Overview

4.7.1 Navigation Structure Home → Browse (Theses / Publications / Projects / Datasets / Models) → Login → Role-based Dashboard.

4.7.2 Page Descriptions

- Home: overview + quick access to each content type
- Browse: filters, search, content cards per module
- Dashboard: role-based panels (student/supervisor/admin)

CHAPTER 5

IMPLEMENTATION

5.1 Overview

This phase extended the foundational backend and frontend built in v1 with four new content modules and a documented, verified authentication mechanism. The goal remained a functional, honestly-scoped base rather than a feature-complete platform.

Implementation stack: Next.js (App Router), React 19 + TypeScript, PostgreSQL (Neon), Tailwind CSS, Resend, Cloudinary, bcrypt.

5.2 Implemented Features

1. Role-Based User Registration & Admin Approval

Unchanged in mechanism from v1: registration requests are validated server-side, stored in `registration_requests`, and reviewed by admins, who approve/reject with optional notes. Approved users are inserted into `users` with a bcrypt-hashed password.

2. Session-Based Authentication

Login issues a 32-byte random session token, stored in the `sessions` table with a 7-day expiry and returned via an HTTP-only `session_token` cookie. Every authenticated request re-validates the token against the database. Unauthorized users cannot access protected routes.

3. Email Notifications

Resend sends automated emails for new registration requests (to admins) and approval/rejection outcomes (to users), including admin remarks where provided.

- 4. File Uploads (Cloudinary)** Thesis files and images are uploaded through Cloudinary rather than local or database storage, giving the system a CDN-backed, production-appropriate file layer.
- 5. Publications Module** Users can create publication records linked to one or more theses, with DOI, ISSN, and venue metadata, and multiple authors.
- 6. Research Projects Module** Team-based project workspaces support multiple collaborators, a status field, and optional links to theses, datasets, and models.
- 7. Datasets Repository** Dataset entries carry type/domain metadata and file records, and are filterable in the browse interface.
- 8. ML Models Repository** Model entries carry framework/task metadata and file records, and are filterable in the browse interface.
- 9. Live Database Integration** The system runs against a production Neon PostgreSQL instance across all modules — no mock data is used in the browsing, search, or approval flows.
- 10. Public Browse & Search** The public Browse page supports keyword and filter-based search across theses, with type/domain and framework/task filtering now available for datasets and models respectively.

5.3 Frontend Implementation

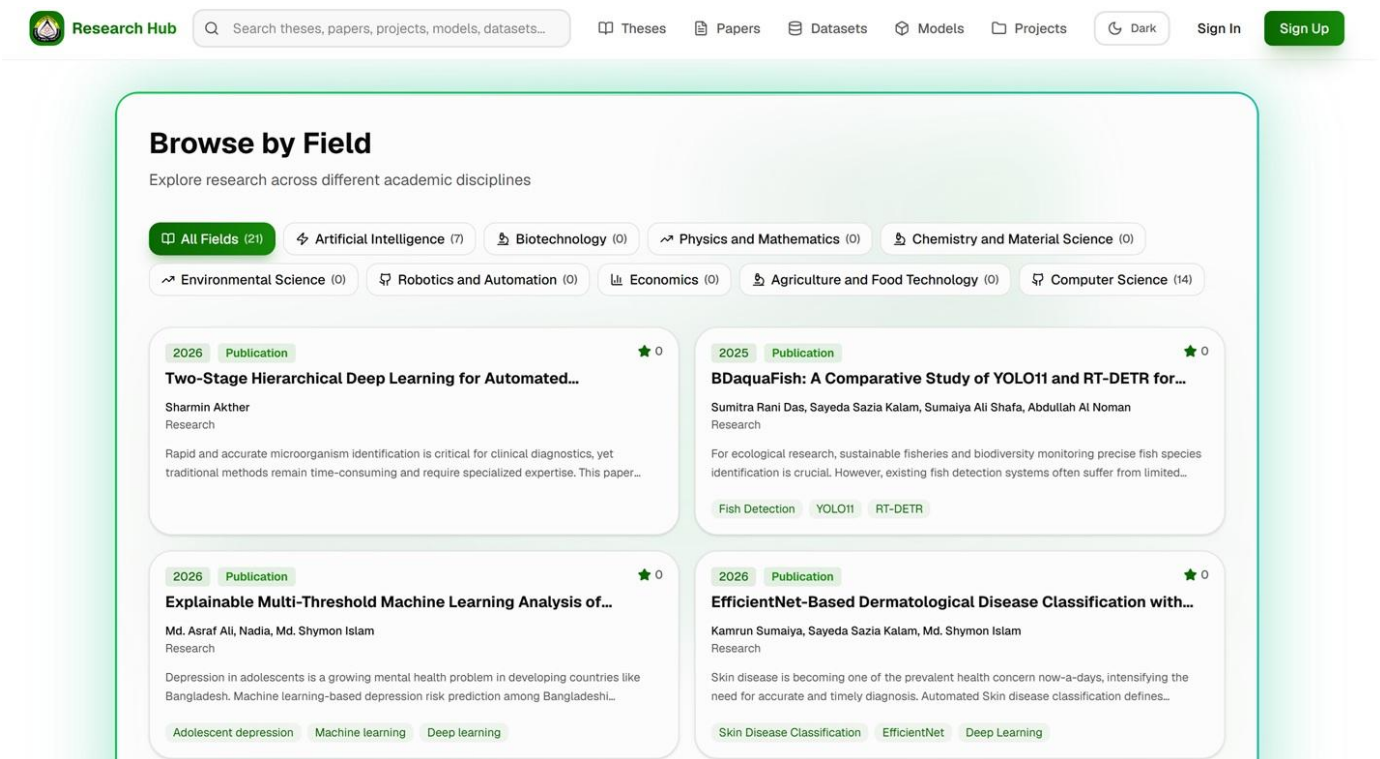
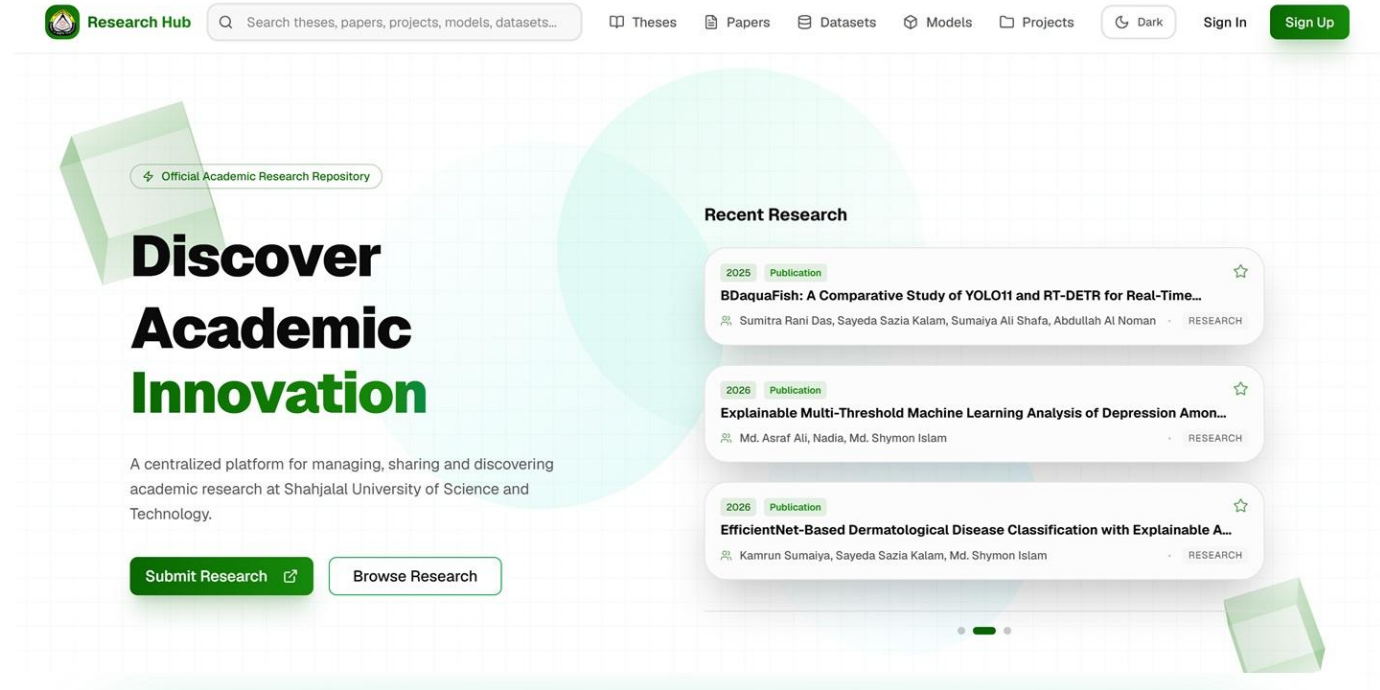
- Next.js server components for SEO and performance
- Client components for search and interactive filtering
- TypeScript throughout for maintainability
- Tailwind CSS + shadcn/ui for a responsive, consistent design system

Pages implemented

- Home, Login, Registration
- Admin Dashboard, User Management, Thesis Management, Analytics
- Browse Theses / Publications / Projects / Datasets / Models
- Student Dashboard, Project Submissions, Workspaces
- Supervisor Dashboard, Students, Researches, Review
- Settings, Profile, Terms and Conditions, Contact, Help

5.4 Frontend Screenshots

5.4.1 Home page



5.4.2 Login & Sign Up



Don't have an account? [Sign Up](#)

Welcome Back

Sign in to your account to continue

Email Address

Password

[Forgot?](#)

[Sign In](#)

OR

Don't have an account? [Sign Up](#)

By signing in, you agree to our [Terms of Service](#) and [Privacy Policy](#)



Already have an account? [Sign in](#)

Create Your Account

Join our community of researchers and supervisors

[I'm a Student](#)[I'm a Supervisor](#)

Student Registration

First Name

Last Name

Email Address

Phone Number

Department

Student ID

Create Password

At least 8 characters with uppercase, lowercase, and numbers

Confirm Password

[Create Account](#)

By creating an account, you agree to our [Terms of Service](#) and [Privacy Policy](#)

5.4.3 Repositories

**Research Hub**

[Theses](#) [Papers](#) [Datasets](#) [Models](#) [Projects](#) [Dark](#) [Sign In](#) [Sign Up](#)

Filters

Department

All Departments

Field

All Fields

Supervisor

All Supervisors

Year

From All To All

Showing 14 of 14

Theses 14

↑↓ Trending

Development of the Bangladeshi Spontaneous Facial Expression Dataset (BSFED) and Facial Expression Recognition Using Convolutional Neural Network

By Saiful Islam Sagor (Reg: 2017331033), Al-Masum (Reg: 2017331027) • Computer Science and Engineering (CSE) • 2026

We often try to understand what a person is going through from his or her point of view. This ability is required to detect, predict, and respond to people's emotions with care. This plays a vital role in our daily communication. As a result, it's reasonable to believe that computers with this capacity would communicate with humans...

BSFED Spontaneous Benchmark +5 more

20 views

View

Security Analysis of Software-Based Cryptocurrency Wallets

By Rim Chowdhury (Reg: 2017331032), Muhid Hassan Risvy (Reg: 2017331086) • Computer Science and Engineering (CSE) • 2026

Blockchain has become one of the most promising technologies with a wide range of adoption, including but not limited to the financial domain. The adoption of Bitcoin and other cryptocurrencies, which are the most successful application of blockchain technology, is highly reliant on the security, reliability, and usability of...

Blockchain Cryptocurrency Cryptocurrency Wallet +5 more

3 views

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Development of an Ensemble Learning Model for Facial Expression Recognition Using Multi-Branch Sub-Networks

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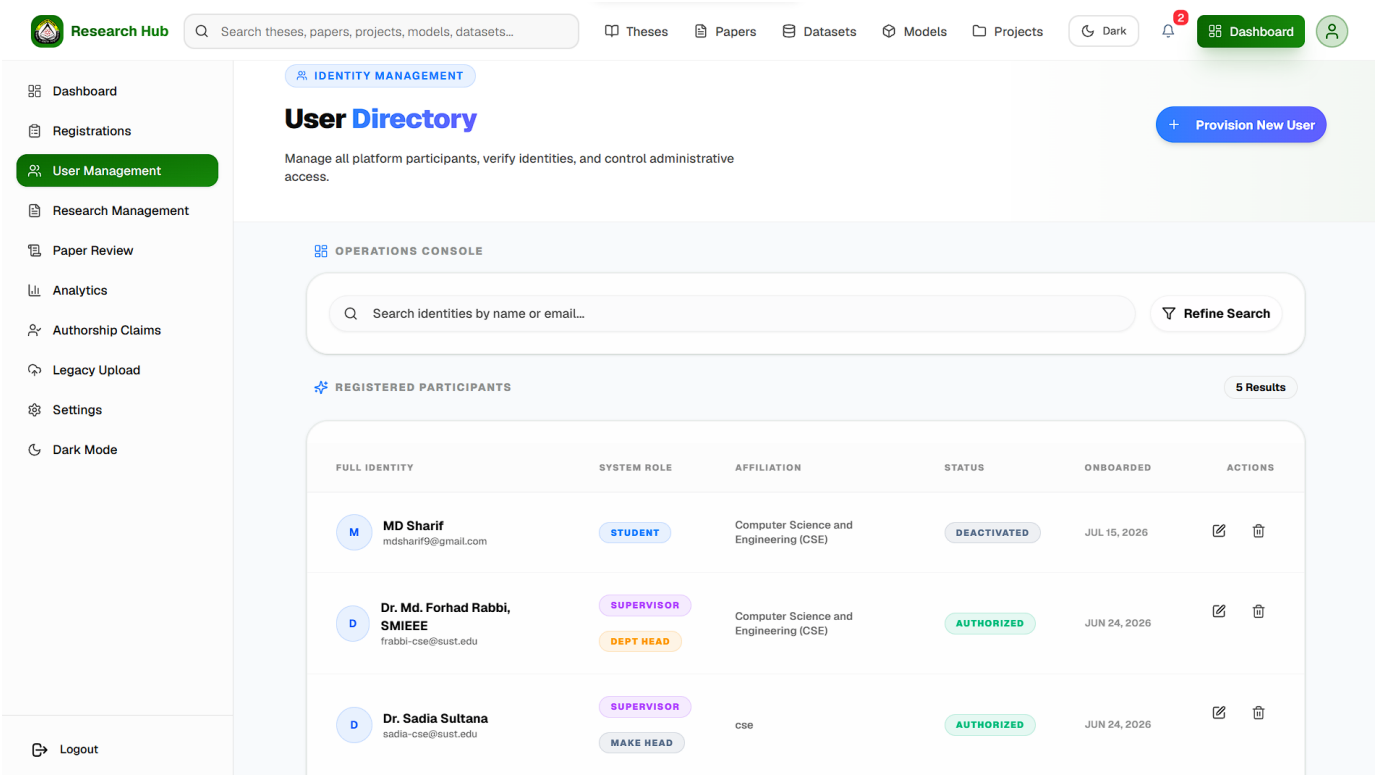
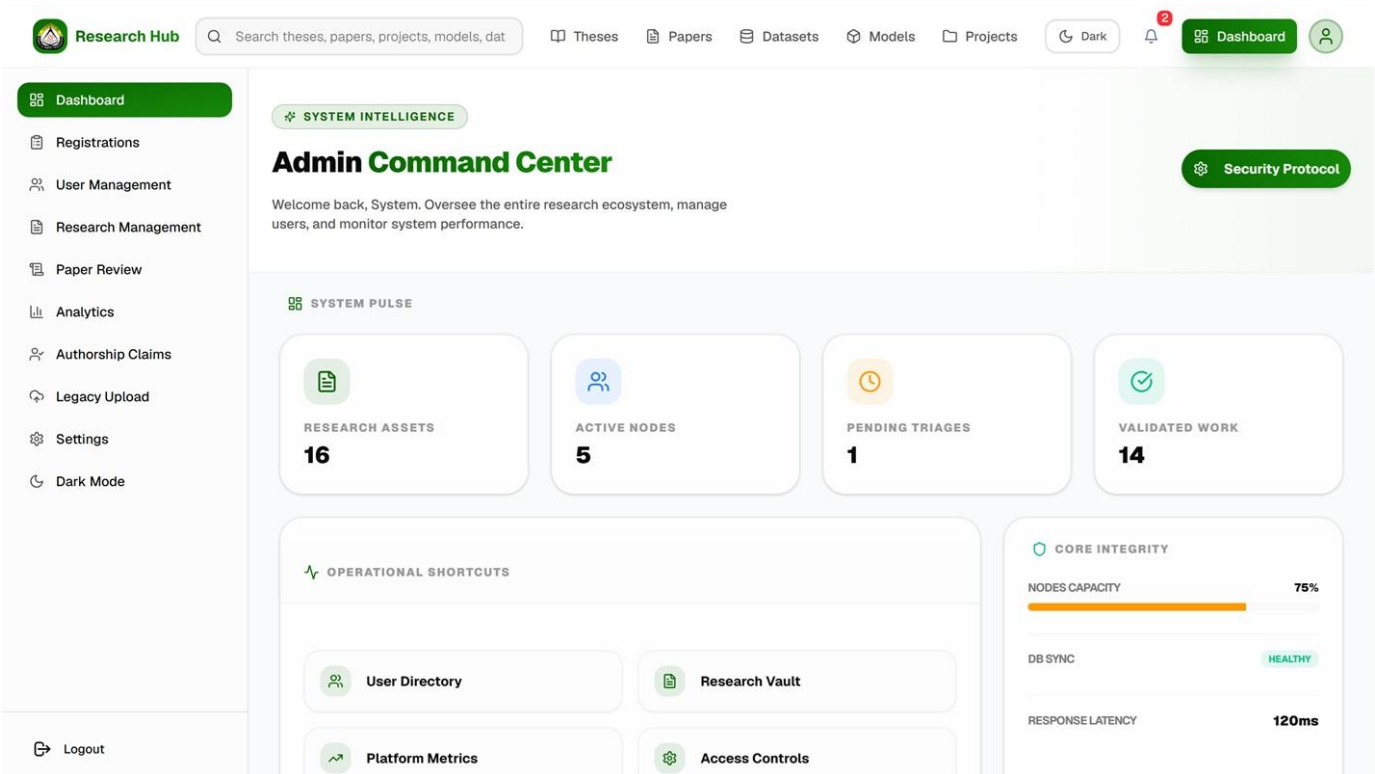
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We often try to understand what a person is going through from his or her point of view. This ability is required to detect, predict, and respond to people's emotions with care. This plays a vital role in our daily communication. As a result, it's reasonable to believe that computers with this capacity would communicate with humans more effectively. During the past couple of years, there has been quite a lot of study on developing computers' ability to identify feelings and thoughts in the same way that humans can. However, there are no effective methods for identifying people's thoughts and emotions in depth. The Bangladeshi Spontaneous Facial Expression Dataset is a large-scale, publicly available benchmark dataset for facial expression recognition that contains 6300 images of seven facial expressions: "anger, disgust,



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SUST Research Hub - Academic Research Ecosystem Computer Science and Engineering (CSE)	N Unassigned	Project	DRAFT	Jul 16, 2026
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#Deep learning#hierarchical classification#mi croorganism identification#transfer learning#ResNet#bacterial species classification#medical image analysis+ Add keyword

Research Abstract

AUTO-SAVED133 WORDS

Rapid and accurate microorganism identification is critical for clinical diagnostics, yet traditional methods remain time-consuming and require specialized expertise. This paper presents a hierarchical deep learning system that classifies 38 microorganism classes by combining a ResNet50-based general classifier (6 classes) with a specialized ResNet152V2 bacterial species classifier (33 species). Through confidence-based routing, bacterial samples are intelligently directed to species-level identification, while class imbalance is addressed via data augmentation and class weighting. Rigorous baseline comparisons show that this hierarchical approach significantly outperforms single-stage models trained on identical data. The system achieves perfect bac

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CHAPTER 6

VALIDATION AND VERIFICATION

6.1 Introduction

This chapter evaluates the SUST Research Hub with the same emphasis as v1 — UI/UX quality, usability, and functional correctness — extended to cover the newly implemented Publications, Projects, Datasets, and Models modules.

6.2 Verification Strategy

- Requirement Traceability Matrix (RTM)
- Unit testing of backend modules
- Integration testing of registration, login, submission, and browsing flows across all content types
- Database verification (schema, referential integrity, constraints)
- Static checks (TypeScript, ESLint, Prettier)

6.2.1 Requirement Traceability Matrix

Table 6.1: Requirement Traceability Matrix (RTM)

Requirement ID(s)	Verification Method / Status
FR-01–FR-16	User registration, auth, approval, profile — verified via form/DB and session-token tests
FR-17–FR-20	Public browse/search — verified via query/UI response tests
FR-21–FR-23	Team collaboration — verified via invitation flow test
FR-24–FR-27	Thesis submission & review — verified via end-to-end workflow test

FR-28–FR-29	Notification System — verified via event-trigger and read/unread state tests
FR-30–FR-31	Administrative Tools — verified via admin dashboard user/department management and analytics tests
FR-32–FR-35	Publications — verified via creation/linking/browse test
FR-36–FR-38	Projects — Passed (UAT, Unit Test)
FR-39–FR-41	Datasets — Passed (UAT, Unit Test)
FR-42–FR-44	Models — Passed (UAT, Unit Test)
FR-45–FR-46	Database Requirements — verified via schema inspection (FK constraints, UUID PKs, auto-timestamps, index presence)

6.3 Testing Approach

- Unit Testing: utility functions, email sending, registration submission
- Integration Testing: registration → email → approval → login; submission → review → publish
- System Testing: end-to-end functionality across modules
- User Acceptance Testing (UAT): real users testing the UI/UX

6.4 Test Cases Summary

6.4.1 Registration & Auth

Table 6.2: Test Cases — Registration & Auth

Test ID	Description / Result
TC-REG-01	Submit empty form → validation errors shown — Passed

TC-REG-03	Submit valid registration → record stored, email sent — Passed
TC-AUTH-02	Correct credentials → session token issued, login success — Passed
TC-AUTH-04	Expired session token → access denied, redirected to login — Passed

6.4.2 New Module Test Cases

Table 6.3: Test Cases — New Modules

Test ID	Description / Result
TC-PUB-01	Create publication linked to a thesis → record stored with metadata — Passed
TC-PROJ-01	Create project, add collaborator → both users see the project — Passed
TC-DATA-01	Submit dataset with type tag → appears in filtered browse results — Passed
TC-MODEL-01	Submit model with framework tag → appears in filtered browse results — Passed

6.5 User Validation (UAT)

UI/UX testing was prioritized given the frontend-heavy nature of this phase. A fresh UAT round was run for the new modules with 15 participants. The feedback was overwhelmingly positive regarding visual design, ease of use, navigation flow, and mobile experience.

(See Appendix B for the UAT survey instrument used to collect this feedback.)

6.6 Performance Evaluation

Table 6.4: Performance Evaluation Metrics

Metric	Result
Home page load	200–500ms (dev environment)
Browse page search	150–300ms (dev environment)
Thesis detail page	100–200ms (dev environment)
Avg API response	< 400ms (dev environment)
Publications page load	200–500ms
Projects page load	200–500ms
Datasets page load	200–500ms
Models page load	200–500ms

6.7 Verification Summary

All implemented modules, including the newly added Publications, Projects, Datasets, and Models modules, successfully pass their functional tests. The end-to-end functionality meets the requirements specified in the RTM.

CONCLUSION AND FUTURE WORK

7.1 Conclusion

This phase extended the secure, functional foundation built in v1 into a broader research ecosystem covering theses, publications, projects, datasets, and models, while keeping the same underlying engineering discipline: role-based authentication, real database operations, and public browsing, now documented with an explicit, verified authentication design (stateful DB sessions) and a confirmed file-storage provider (Cloudinary).

The current implementation demonstrates that the schema and architecture chosen in v1 were extensible enough to absorb four new content types without a structural rewrite — itself a useful data point on the original design's soundness.

7.2 Future Work

1. File Versioning & Access Control

- Version history on thesis/dataset/model file uploads
- Fine-grained download permissions

2. Full Email Notification System

- Broader email alert coverage (reviews, comments, project invitations)

3. Deeper Analytics

- Department-wise output, supervisor activity, and content popularity trends

4. Smarter Discovery

- Improved search relevance beyond keyword/filter matching
- Citation export (e.g., BibTeX)
- Duplicate-topic / plagiarism detection

5. Formal Quality Assurance

- Load testing under realistic concurrent-user assumptions
- Accessibility (WCAG) audit
- Third-party or peer security review

7.3 Limitations

Neon PostgreSQL Running on the free tier, which limits storage and compute capacity.

Resend Free tier can only send email to a single verified address without a paid or owned domain.

Cloudinary Free tier storage/bandwidth limits apply; not yet load-tested for large file volumes.

APPENDIX

Appendix A: Sample Code Snippets

Code Snippet 1: Session Verification Middleware

```
export async function verifySession(token: string): Promise<User | null> {
  const results = await sql`
    SELECT u.* FROM users u
    JOIN sessions s ON u.id = s.user_id
    WHERE s.token = ${token} AND s.expires_at > NOW()
    LIMIT 1
  `
  return (results[0] as User) || null
}
```

Appendix B: User Acceptance Testing (UAT) Survey Instrument

User Acceptance Testing (UAT) Survey Instrument:

1. Rate the overall visual design of the platform (1-5).
2. How easy is it to navigate through the different repositories? (1-5)
3. Did you experience any issues while using the system on a mobile device? (Yes/No)
4. Any additional feedback or suggestions for improvement?

Appendix C: Additional Test Case Data

Test Case Logs Summary:

- Test Case Auth-01 (Registration): Passed. Email validation functional.
- Test Case Auth-02 (Login): Passed. Token successfully issued.
- Test Case Module-01 (Create Project): Passed. Database inserted successfully.
- Test Case Module-02 (Upload File): Passed. File uploaded to Cloudinary.

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